ITCS285: Database Management Systems
Project description,
May 17th, 2026, Weight 15%

The objective of this project is to enable students to develop a working relational DB using necessary tools. Students will be able to learn DB design methodology for relational DB and create a DB which can store data efficiently, retrieve information effectively, and free from data anomalies. The DB development process consists of the following steps:

1. Collect and describe the requirements details of your selected application.
2. Draw the appropriate ER/EER representation of your application.
3. Design the relational scheme of your application based on your ER/EER representation. All constraints (Keys, Integrity, Referential, …) should be clearly stated.
4. Select a DBMS System (Oracle, MySQL, SQL Server, online SQL interpreters…) to create your relations schemas and populate them with at least 20 tuples for each relation.
5. Design different SQL queries (at least 10 covering the different features of SQL). Run these queries on the selected DBMS and present the results. Your queries should involve most of the relations you designed and include complex queries. They should consider Multiple tables join; Use sub-queries; Group data using GROUP BY; Sort query results using ORDER BY; IN or NOT IN; LIKE; Aggregate functions or calculated fields.
6. Contribution sheet: attach a sheet that shows your meetings and contribution of each member (weekly meetings are recommended).

Students are supposed to select one from the list of project ideas given below. **Any topic different from the proposed topics will be rejected**. The proposed ideas are:

1. Online Examination Management System
2. Expense Tracker System
3. Tour Package Organizer
4. Alumni Database
5. Research Publication Tracker
6. Fraud Detection Management System

Project group should include **3-4 students** from **the same section**. The group's details and the selected idea should be sent to your instructor no later than **April 16th, at 6pm**. The deadline of the project is due **May 17th, 6pm**. The project expected outputs are:

1. The project report **(15 to 20 pages maximum)** includes details of the project steps.
2. Students should consider the followings points:
   - **Plagiarism is strictly prohibited**. Groups caught copying each other's reports or copying past semesters' projects or downloading ready projects/reports will be rejected and receive negative marks.
   - **Collection of data** (if any) should be done through well-known methods such as surveys or official requests for data from organizations (in such cases, a supporting letter can be provided by the department). Students can also use datasets repositories available on the Internet.
   - **Conflicts.** Any conflict between group members or related to any other issue should be reported to your instructor as soon as possible.
   - **Grading.** The grading scheme will be mainly based on elements 1 to 6 of this proposal.